

Transmission electron microscopy (TEM):

- TEM is also known as conventional transmission electron microscopy or CTEM.
- Max Knoll and Ernst Ruska invented it in 1933 in Berlin.
- Recent electron microscopy (based on transmission) commonly contains a beam column which is around 2.5m tall and has a 30cm diameter, and has an ability to attain a $2\text{\AA}$ resolution.
- This technique is utilized for analysing the surface structure, i.e., morphology, surface imperfection, i.e., defects, crystal structure of the atom, size of the particle and also sample's composition.

Design of TEM is similar to a light microscope-

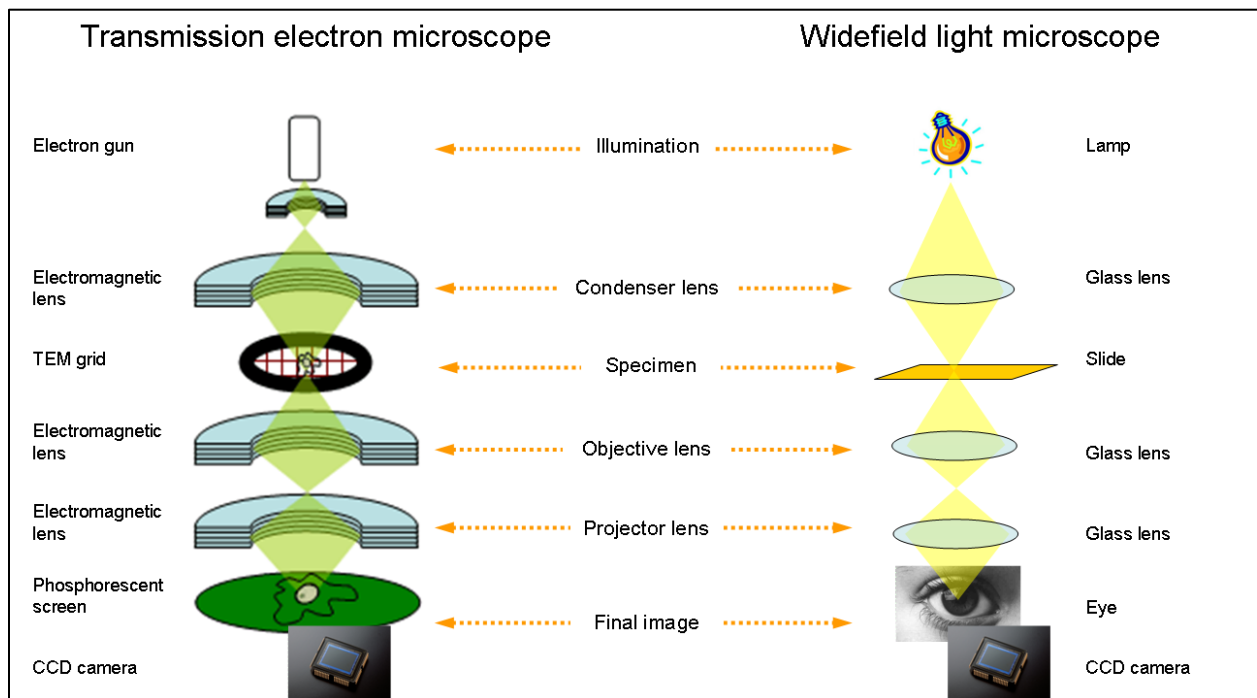


Figure 1 Similarity of a transmission electron microscope with a wide field light microscope.

Working Principle-

An extremely thin sample is required for scanning in TEM from which electron beam is passed through rendering its interaction with the sample as a result of which image is produced. This image can be magnified and focused on the device used for imaging, like a fluorescent screen, on a photographic film layer, or to be identified by a sensor like a CCD camera.

Instrumentation of TEM-

- Source of electron
- Gun based on Thermionic Emission

- Beam of Electron
- Electromagnetic lenses
- Vacuum chamber
- Two Condensers lenses, objective and intermediate lens
- Sample holder and stage
- (Imaging Device) Phosphor or fluorescent screen
- Computer

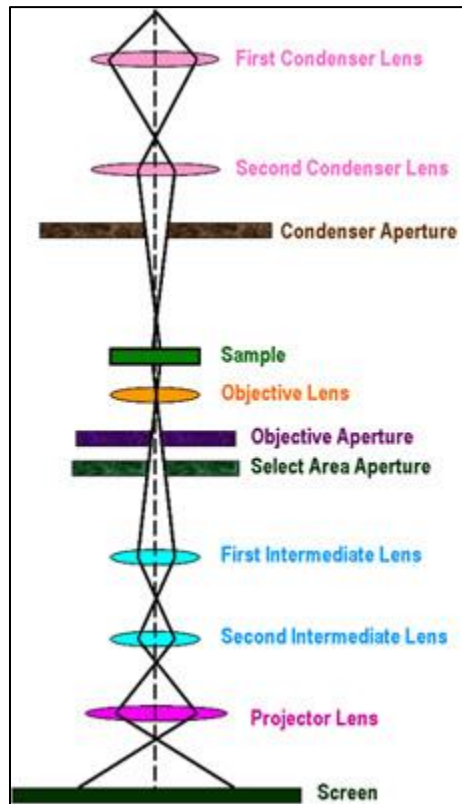


Figure 2 Instrumentation of TEM.

Electron Gun:

Electrons can be produced either by thermionic emission or by a process called cold field emission. During thermionic emission, a very fine tip of a tungsten filament, a LaB₆ crystal or a ZrO/W Schottky emitter, is heated by an electrical current flowing through it, thereby enabling the emission of electrons. The electrons leaving the filament have a low energy and, therefore, need to be accelerated to the desired speed before entering the electron column. A high voltage between the electron source (cathode) and an anode plate is applied leading to an electrostatic field through which the electrons are guided and accelerated. During cold field emission, the electrons can escape from an extremely fine tungsten tip without heating (at room temperature). The advantage of cold field emission sources is the very high yield of electrons and the very low chromatic aberration of the electrons allowing imaging at atomic resolutions. These instruments are very costly and require very high vacuum.

The Electron gun working can be controlled based on 3 parameters:

- The accelerating voltage,
- Current of the filament (and therefore its temperature),
- And the Wehnelt cap bias voltage.

The temperature of the filament tip is controlled by the filament current which in turn controls the amount of emitted electrons. The filament current is increased till the number of emitted electrons no longer increases, which actually means that filament is saturated in order to maximize the emission.

The passing current between the system having high voltage and ground is controlled by bias resistor setting which in turn is controlled by the gun bias. When small bias voltage is used, the wehnelt negative potential is ineffective in comparison to the filament, leading to poor focusing of the electrons that are accelerated toward the anode. This results in beam spreading, causing it to appear weak on the screen. When the biasing is increased, the focusing action is improved therefore the effective beam brightness is also increased; but, beyond a certain value the Wehnelt is so negative in comparison to the filament that the brightness starts to decrease because electrons are not permitted to emit from the filament or, in a case they are emitted, they are repelled back in the direction of the filament.

The point at which the finest brightness of the beam is attained is determined by the distance between the Wehnelt and filament.

Electromagnetic lenses:

Electromagnetic lenses consist of a huge bundle of windings of insulated copper wire, a soft iron cast and pole piece (**Figure 3A**). A magnetic field is induced by the current in the winding and reaches its main strength at the pole piece of the lens. The accelerated electrons entering the magnetic field are deviated by Lorentz forces. The direction of both magnetic field as well as electrons defines the resultant force which is always perpendicular to the plane. In conclusion, the electrons take a circular path through the lens system (**Figure 3B**).

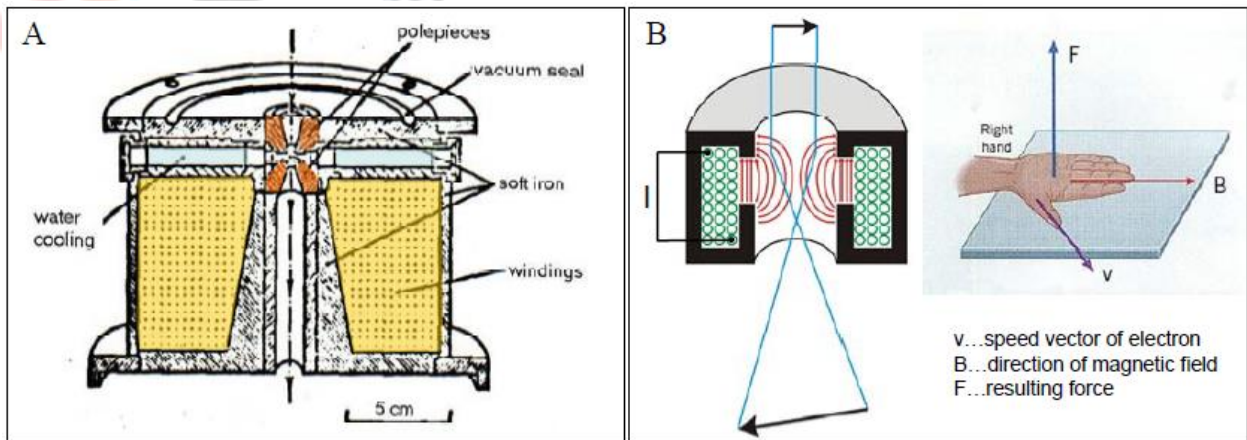


Figure 3 Electromagnetic lens. A) The magnetic field is strongest in the area of the pole piece; B) Electrons passing the magnetic field are deviated perpendicular to the plane defined by the magnetic field B and the velocity vector v.

Condenser lens system:

The beam diameter is reduced and controlled by condenser lens system. The purpose of the first condenser C1 lens (or spot size) which is a strong lens is to de-magnify the electron source image by around $\times 1/100$ to provide a small “point” source at the “crossover” that is more coherent than the large (50 μm diameter) tip of the filament. The purpose of the second condenser C2 lens (brightness or intensity) which is a weaker lens is to project the de-magnified image of the source on top of the sample by a magnification of $\times 2$, giving an overall demagnification of $\times 1/50$. Illumination spread onto the screen is controlled by this lens. A part named condenser aperture is positioned just below or sometimes between the condenser lenses; its role is to collimate (i.e. making parallel) the beam of the electron as well as modification in its intensity.

Objective and intermediate lenses:

The reason behind the back focal plane being very close to the lens itself is because the magnification factor of the objective lens is larger. Aperture of the objective (it is the middle aperture on the column) is mounted in the back-focal plane. The selected area aperture sits in the first image plane below the specimen, which is below both the objective lens and the objective aperture.

By altering the first projector lens excitation (also known as intermediate lens or diffraction lens), either an image or a diffraction pattern is produced.

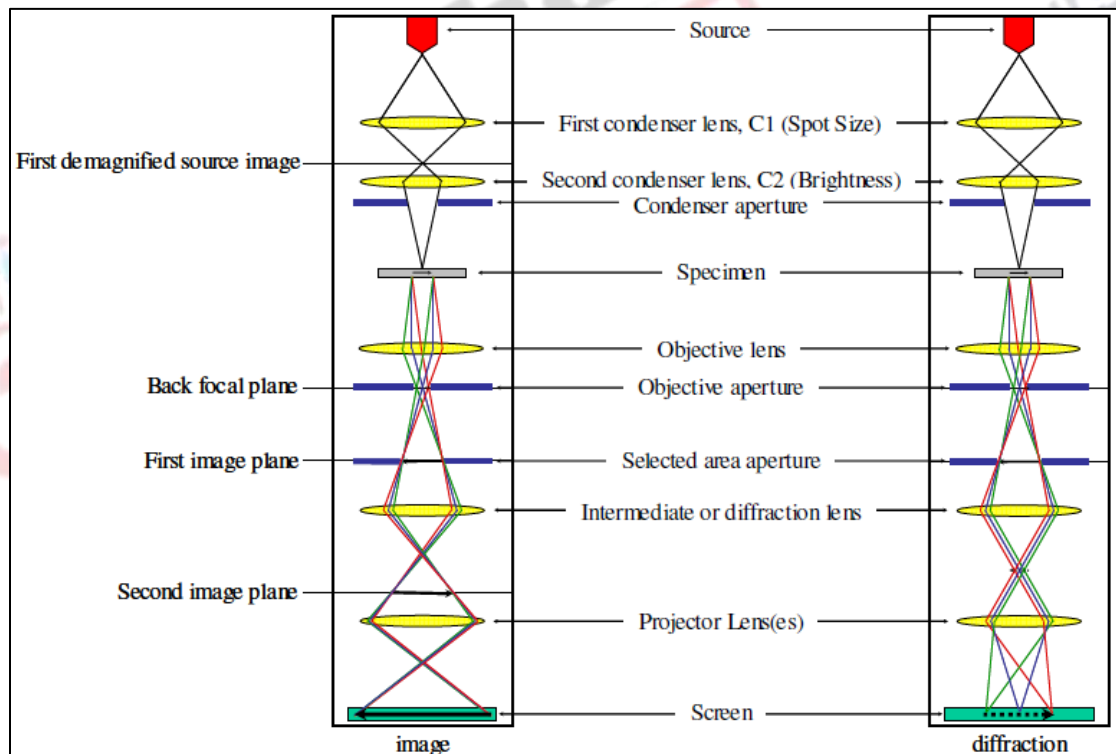


Figure 4 Diagram showing the formation of both an image and a diffraction pattern.

Specimen holders and stages:

In TEM, the electron column does not offer a lot of space for the sample. Further, the sample should be fine (thin) so that the electrons can penetrate the specimen to produce an image. The average thickness of a biological specimen should be around 70 nm for a TEM with an acceleration voltage for the electrons of ~ 100 kV (higher voltages allow the investigation of thicker samples). Thin sections of the sample are

mounted on copper grids of 3 mm diameter, which are available in a wide variety of materials and mesh sizes. The grids with the sections on top are attached in a holder and introduced into the goniometer of the TEM through a vacuum lock, since the system always stays under high vacuum. The goniometer is the mechanical setup which enables highly precise and stable control of the specimen holder during imaging. Any drift or instability results in a blurred image, particularly at high magnifications (**Figure 5**).

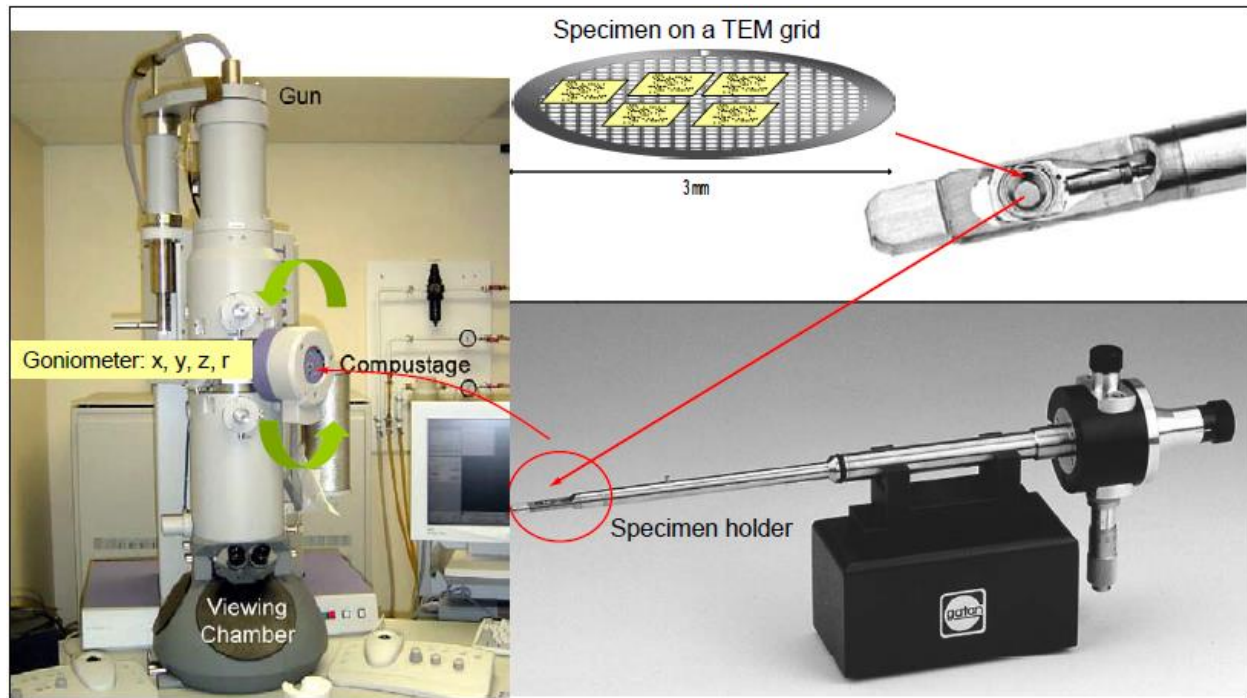


Figure 5 Thin sections of a specimen on a TEM grid, holder tip and complete specimen holder, which is introduced into the goniometer of the TEM through a vacuum lock.

Vacuum System:

Vacuum system is employed in electron microscopes for 4 reasons:

- As electrons are readily scattered, electrons have a mean free path of $\sim 1\text{cm}$ at atmospheric pressure; however, at 10^{-6} Pa they can have mean free path as high as 6.5m .
- The purpose of the vacuum system is to provide insulation between the filament of both anode and cathode as well as in the region around the field emitters, thus hampering undesirable discharge of the electron gun.
- In order to inhibit the oxidation and 'burning out' of the filament, oxygen is eliminated around the filament.
- Samples contamination is decreased by reducing the interaction amongst electron beam and molecules of the gas.

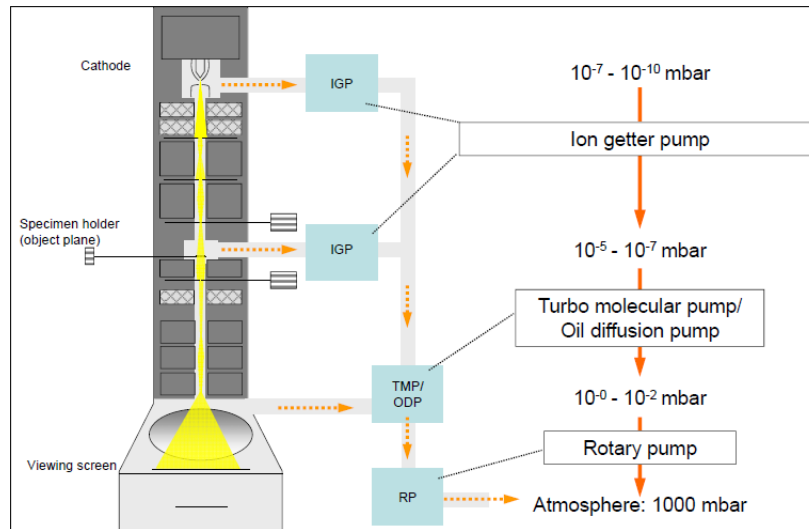


Figure 6A typical vacuum system for a TEM. RP-rotary pump, TMP-turbo molecular pump, IGP-ion getter pump, ODF-oil diffusion pump.

Various microscope parts are differently vacuumed as per their requirements. The gun requires 10^{-9} Pa vacuum, while the specimen requires 10^{-6} Pa and the projection chamber plus camera requires 10^{-5} Pa.

Vacuums can be categorized as: rough (100 - 0.1 Pa), low (10^{-1} - 10^{-4} Pa), high (10^{-4} - 10^{-7} Pa), or ultrahigh ($< 10^{-7}$ Pa).

Phosphor or fluorescent screen (Imaging Device):

There are 2 procedures for specimen observation in TEM as shown in **Figure 4**.

1. Image mode
2. Diffraction mode

In case of image mode, the electron beam hitting the sample is controlled by condenser lens and aperture, the beam which is transmitted will be focused and enlarged by objective and projector lenses, and the image is formed on the screen with identifiable information in relation to the microstructure of the sample. In case of diffraction mode, at the fluorescent screen a diffraction pattern (of electron) is obtained which originates from the electron beam illuminated sample region. The pattern of diffraction is completely similar to that of a pattern of X-ray diffraction. The spot pattern is produced by a single crystal on the screen whereas poly-crystal produces a pattern of powder or ring. The purpose of the image mode is to analyse microstructure, e.g. the grain size, and lattice defects, whereas the use of diffraction mode is to examine crystalline structure.

Image Modes of TEM-

In TEM, the 2 primary image modes vary in the manner of using the objective aperture as filter in electron optics system. These modes are:

1. Bright field microscopy

2. Dark field microscopy

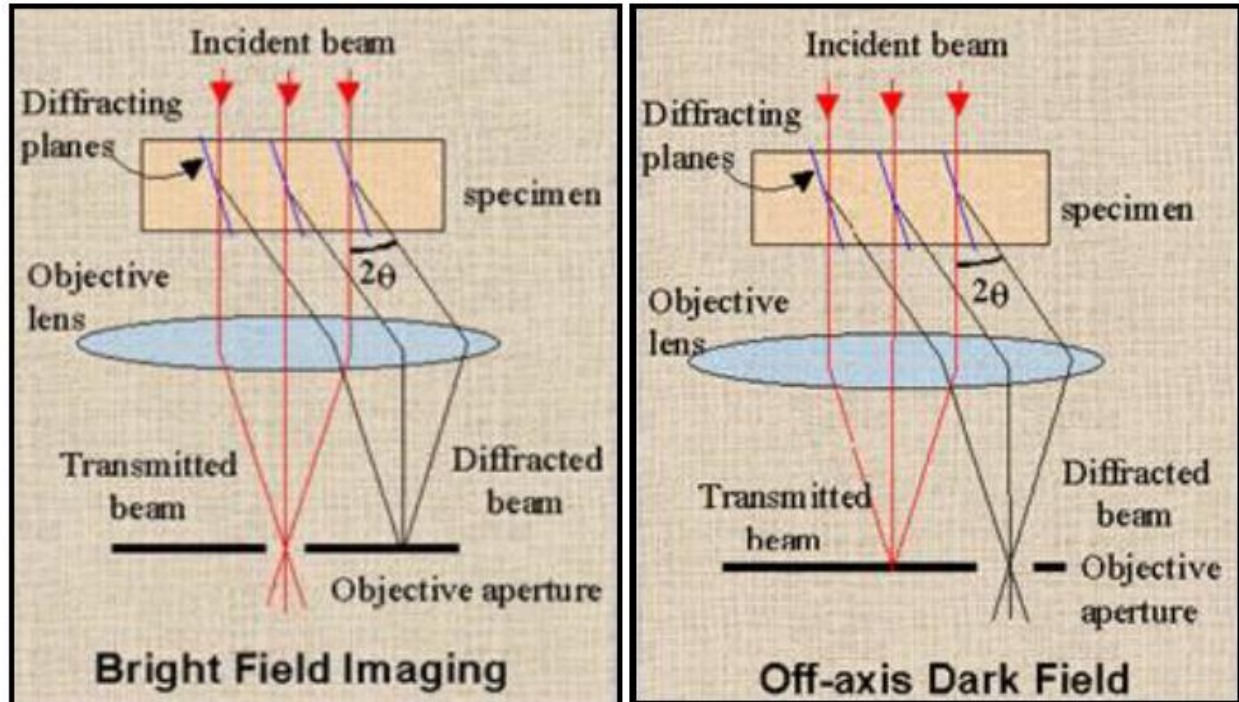


Figure 7 Two image modes of TEM.

In bright field imaging, the image of the sample is created by the electrons that pass through the film without diffracting. An aperture is used to stop the diffracted electrons. In the corresponding dark field imaging mode, the image is formed by the diffracted beam. The technique is called as bright field which is mainly sensitive to extended crystal lattice defects in an otherwise ordered crystal, e.g., dislocations. The electron rays corresponding to bright field and dark field imaging are shown in **Figure 7**.

Electron interaction with matter-

The interaction between the electron beam and the sample is coulombic. The negatively charged electrons can interact strongly with the electron cloud in the solid and also the positively charged nucleus. In contrast, X-rays are EM radiation and they only interact with the electron cloud. In TEM, for imaging purposes, only the forward scattered electrons are of interest. There are two main types of scattered radiation:

- Elastic - this represents coherent scattering (mainly) with no loss of energy. There is also a phase relation with the incident radiation.
- Inelastic - the energy of the scattered electrons is lower than the incident beam. These are also incoherent radiation with no phase relation with the incident radiation.

TEM sample preparation-

Significant part of TEM is its sample preparation for the analysis. There are two main conditions for TEM sample preparation:

- Electron transparent sample must be used. If not the whole sample at least the ROI should be thin. The allowed thickness value for the metallic samples is 30 - 50 nm. Usually, 100 nm is an upper limit for the sample thickness.
- The sample must be mechanically strong for treatment.

TEM samples are either self-supported or mounted on a grid for analysis. Copper grids are the most commonly used, though for high temperature work Mo grids are used. For nanoparticles and thin films a-C film is used as support. A-C has low contrast in the TEM and will not obscure the contrast arising from the specimen. Some typical TEM grids are shown in figure 7.

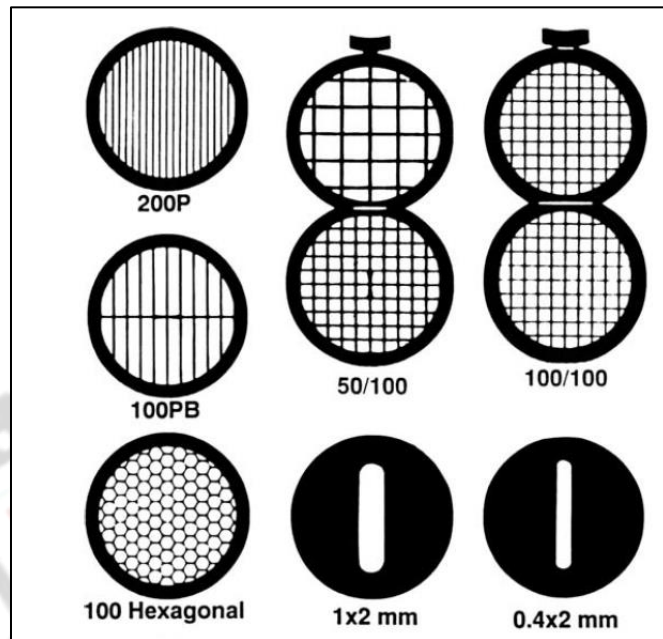


Figure 8 Typical TEM grids.

Transmission Electron Microscope – Principle, Components and Applications

Dr. M. Kannan

Transmission electron microscopy (TEM) is the original form of electron microscopy and analogues to the optical microscope. It can achieve a resolution of ~ 0.1 nm, thousand times better resolution, cannot be reached by the light microscope. The beam of electrons passes through the specimen and analyzes the internal structure of the specimen in the form of images. The electron has the poor penetrating capability and gets absorbed in the thick specimen. Therefore, the thickness of the specimen should not be more than few hundred Angstroms (one angstrom = 10^{-10} m). However sometimes, slightly thickens samples are used in High Voltage Electron Microscope.

Components of TEM

Transmission electron microscope has three essential systems: (1) an electron gun, which produces the electron beam, and the condenser system, which focuses the beam onto the object (sample), (2) the image-producing system, consisting of the objective lens, movable specimen stage, objective, intermediate and projector lenses, which focus the electrons passing through the specimen to form a real, highly magnified image, and (3) the image-recording system, which converts the electron image into some form perceptible to the human eye. The image-recording system usually consists of a fluorescent screen for viewing and focusing the image and a digital CCD camera for permanent records. In addition, a vacuum system, consisting of pumps and their associated gauges and valves, and power supplies are required.

Illumination system (Electron gun and condenser lenses)

Electron gun

The source of electrons, the cathode, is a heated a sharply pointed rod shaped

lanthanum hexaboride. The filament is surrounded by a control grid called as Wehnelt cylinder, with a central aperture arranged on the axis of the column; the apex of the cathode is arranged to lie at or just above or below this aperture. The cathode and control grid are at a negative potential equal to the desired accelerating voltage and are insulated from the rest of the instrument. The final electrode of the electron gun is the anode, which takes the form of a disk with an axial hole. Electrons leave the cathode and accelerate toward the anode. The control and alignment of the electron gun are critical in ensuring satisfactory operation.

Condenser lenses system

The intensity and angular aperture of the beam are controlled by the condenser lens system between the gun and the specimen. A single lens may be used to converge the beam onto the object, but, more commonly, a double condenser is employed. In this the first lens is strong and produces a reduced image of the source, which is then imaged by the second lens onto the object. The use of a small spot size minimizes disturbances in the specimen due to heating and irradiation.

The image-producing system

Objective lenses and projector lenses

The specimen grid is carried in a small holder in a movable specimen stage. The objective lens is usually of short focal length (1–5 mm) and produces a real intermediate image that is further magnified by the projector lens or lenses. A single projector lens may provide a range of magnification of 5:1, and by the use of interchangeable pole pieces in the projector a wider range of magnifications may be obtained. Modern instruments employ two projector lenses (one called the intermediate lens) to permit a greater range of magnification and to provide a greater overall magnification without a commensurate increase in the physical length of the column of the microscope.

For image stability and brightness, the microscope is often operated to give a final magnification of 1,000–250,000 $\times$ on the screen. If a higher final magnification is required, it may be obtained by photographic or digital enlargement. The quality of the final image in the electron microscope depends largely upon the accuracy of the various mechanical and electrical adjustments with which the various lenses are aligned to one another and to the illuminating system. The lenses require power supplies of a high degree of stability; for the highest standard of resolution, electronic stabilization to better than one part in a million is necessary. The control of a modern electron microscope is carried out by a computer, and dedicated software is readily available.

Image translation system (Fluorescent screen and Digital photographic unit)

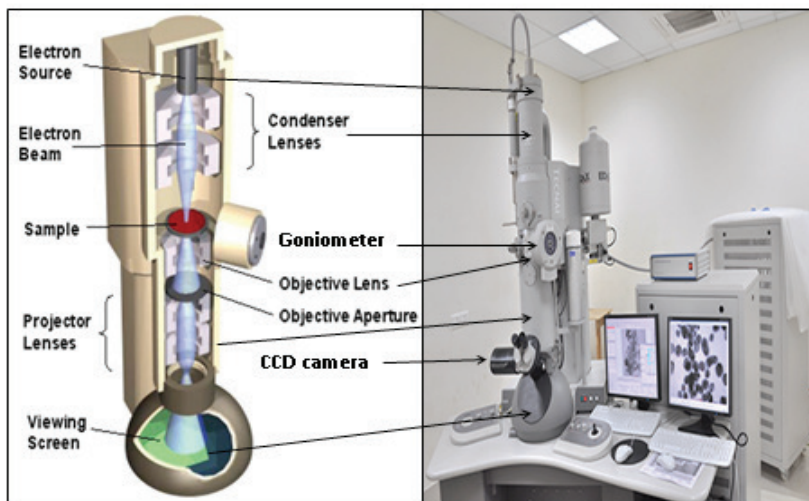
TEM provides informations in the form of variations of electron intensity in the image. The electron image is monochromatic and must be made visible to the eye either by allowing the electrons to fall on a fluorescent screen fitted at the base of the

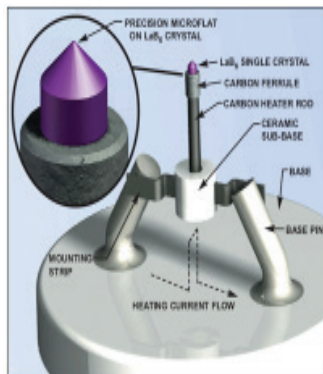
microscope column or by capturing the image digitally for display on a computer monitor. Computerized images are stored in a format such as TIFF or JPEG.

How Transmission electron microscopy (TEM) works

TEM involves a high voltage electron beam emitted from a tungsten filament (cathode) by electrical heating; the shaft of electrons beam drawn toward an anode (magnetic lenses) and pass through an aperture. The beam traverses the aperture and next moves through an electromagnetic condenser, objective, intermediate and projector lens. The focused electron beam transmitted through a very thin specimen (50nm size, semitransparent for electrons and carries information about the structure of the specimen) loaded on a grid inserted in the path and manipulated by goniometer. The part of the beam absorbed scattered and transmitted through objective aperture and projected by projector lens after corrected by intermediate lenses on the fluorescence screen. The image is observed with the help of optical binocular attached to its viewing window. The spatial variation in the “image” is then magnified by a series of magnetic lenses until it is recorded by hitting a fluorescent screen or light sensitive sensor such as a CCD (charge-coupled device) camera fitted either in the side or bottom of the photographic plate. TEM electron scattering rather than differences in absorbance produces contrast in the image. Scattering results from an interaction between specimen atoms and electrons of the illuminating beam. The negatively charged electron clouds around atomic nuclei scatter electrons by repelling them. These effects increase as electrons pass closer to specimen atoms. Nuclei of high atomic number, as in atom of heavy metals such as lead and uranium, cause a widest scattering. Transmission electron microscopes produce two-dimensional, black and white images. The TEM can easily resolve structure such as ribosome, microtubules, microfilaments and large molecules such as proteins. Even images of individual heavy metal atom have been produced under the special operating conditions.

Transmission Electron Microscopy (TEM)

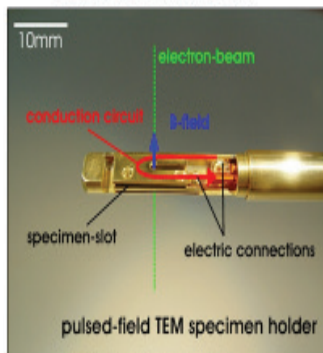




Electron gun source LB 6



Electromagnetic lens



TEM specimen holder



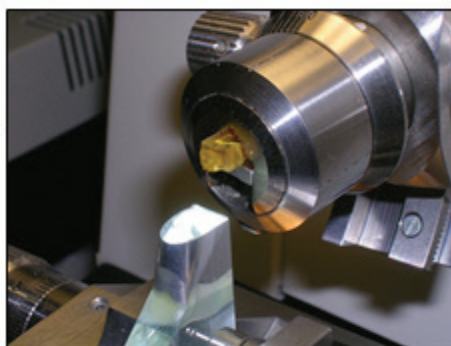
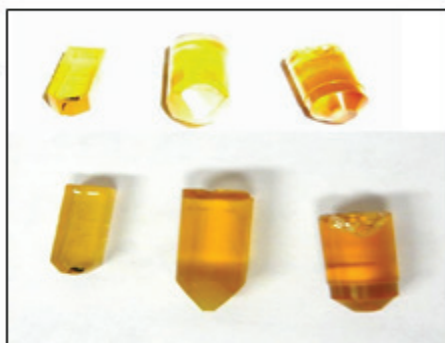
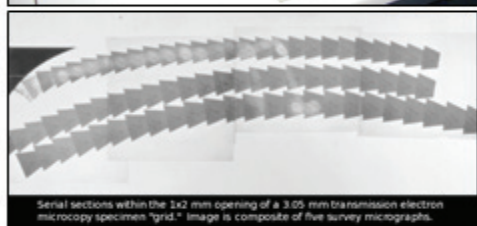
TEM Copper grid mesh

Biological sample preparation techniques for TEM

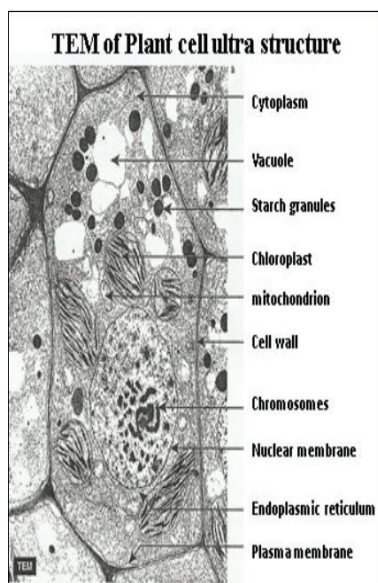
1. Tissue isolation
2. Fixation with glutaraldehyde, OsO_4 and occasionally KMnO_4
3. Embedding in a plastic resin
4. Ultramicrotomy
5. Post-staining of thin sections
6. Photography

Ultramicrotomy of processed biological samples

Ultramicrotome



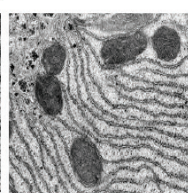
TEM micrographs of plant cell organelles



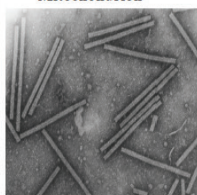
Mitochondrion



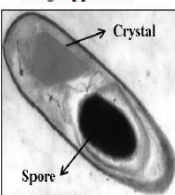
Golgi apparatus



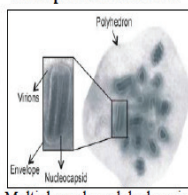
Endoplasmic reticulum



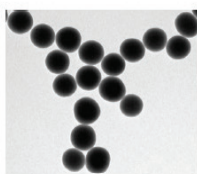
Tobacco mosaic virus



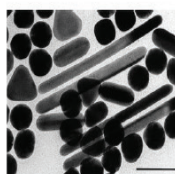
Bacillus thuringiensis



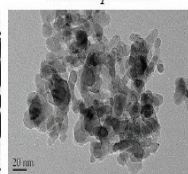
Multiple nucleopolyhedrovirus of Rachiplusia ni



Ag Nano particle



Au Nano particle

SiO₂ Nano particle

Difference between TEM and SEM

Sl. No.	Feature	TEM	SEM
1.	Electron Beam	Broad, static beams	Beam focused to fine point; sample is scanned line by line
2.	Voltages Needed	Accelerating voltage much lower; not necessary to penetrate the specimen	SEM voltage ranges from 60-300,000 volts
3.	Interaction of the beam electrons	Specimen must be very thin	Wide range of specimens allowed; simplifies sample preparation
4.	Imaging	Electrons must pass through and be transmitted by the specimen	Information needed is collected near the surface of the specimen
5.	Image Rendering	Transmitted electrons are collectively focused by the objective lens and magnified to create a real image	Beam is scanned along the surface of the sample to build up the image

Application of TEM

- Transmission Electron Microscope is ideal for a number of different fields such as life sciences, nanotechnology, medical, biological and material research, forensic analysis, gemology and metallurgy as well as industry and education.
- TEMs provide topographical, morphological, compositional and crystalline information.
- The images allow researchers to view samples on a molecular level, making it possible to analyze structure and texture.
- This information is useful in the study of crystals and metals, but also has industrial applications.
- TEMs can be used in semiconductor analysis and production and the manufacturing of computer and silicon chips.

- Technology companies use TEMs to identify flaws, fractures and damages to micro-sized objects; this data can help fix problems and/or help to make a more durable, efficient product.
- Colleges and universities can utilize TEMs for research and studies.
- Although electron microscopes require specialized training, students can assist professors and learn TEM techniques.
- Students will have the opportunity to observe a nano-sized world in incredible depth and detail.
- Advantages
- TEMs offer the most powerful magnification, potentially over one million times or more
- TEMs have a wide-range of applications and can be utilized in a variety of different scientific, educational and industrial fields
- TEMs provide information on element and compound structure
- Images are high-quality and detailed
- TEMs are able to yield information of surface features, shape, size and structure
- They are easy to operate with proper training
- Disadvantages
- Some cons of electron microscopes include:
- TEMs are large and very expensive
- Laborious sample preparation
- Potential artifacts from sample preparation
- Operation and analysis requires special training
- Samples are limited to those that are electron transparent, able to tolerate the vacuum chamber and small enough to fit in the chamber
- TEMs require special housing and maintenance
- Images are black and white
- Electron microscopes are sensitive to vibration and electromagnetic fields and must be housed in an area that isolates them from possible exposure.
- A Transmission Electron Microscope requires constant upkeep including maintaining voltage, currents to the electromagnetic coils and cooling water.